

Name : Belal Mohamed Hassn

ID : 200048003

====Part A=====

```
belal@kali: ~  
Session Actions Edit View Help  
(belal@kali)-[~]  
$ pwd  
/home/belal  
(belal@kali)-[~]  
$ cd /  
(belal@kali)-[/]  
$ ls -l  
total 64  
lrwxrwxrwx 1 root root 7 Sep 3 17:44 bin → usr/bin  
drwxr-xr-x 3 root root 4096 Sep 3 18:38 boot  
drwxr-xr-x 18 root root 3280 Sep 4 12:52 dev  
drwxr-xr-x 185 root root 12288 Sep 4 12:52 etc  
drwxr-xr-x 3 root root 4096 Sep 3 18:33 home  
lrwxrwxrwx 1 root root 34 Sep 3 17:46 initrd.img → boot/initrd.img-6.19.14+kali-amd64  
lrwxrwxrwx 1 root root 34 Sep 3 17:46 initrd.img.old → boot/initrd.img-6.19.14+kali-amd64  
lrwxrwxrwx 1 root root 7 Sep 3 17:44 lib → usr/lib  
lrwxrwxrwx 1 root root 9 Sep 3 18:00 lib32 → usr/lib32  
lrwxrwxrwx 1 root root 9 Sep 3 17:44 lib64 → usr/lib64  
drwx----- 2 root root 16384 Sep 3 17:44 lost+found  
drwxr-xr-x 3 root root 4096 Sep 3 17:44 media  
drwxr-xr-x 2 root root 4096 Sep 3 17:45 mnt  
drwxr-xr-x 3 root root 4096 Sep 3 18:00 opt  
dr-xr-xr-x 363 root root 0 Sep 4 12:52 proc  
drwx----- 4 root root 4096 Sep 3 18:39 root  
drwxr-xr-x 43 root root 980 Sep 4 12:52 run  
lrwxrwxrwx 1 root root 8 Sep 3 17:44/sbin → usr/sbin  
drwxr-xr-x 3 root root 4096 Sep 3 18:05 srv  
dr-xr-xr-x 13 root root 0 Sep 4 14:05 sys  
drwxrwxrwt 14 root root 340 Sep 4 14:20 tmp  
drwxr-xr-x 15 root root 4096 Sep 3 18:00 usr  
drwxr-xr-x 12 root root 4096 Sep 3 18:39 var  
lrwxrwxrwx 1 root root 31 Sep 3 17:46 vmlinuz → boot/vmlinuz-6.19.14+kali-amd64  
lrwxrwxrwx 1 root root 31 Sep 3 17:46 vmlinuz.old → boot/vmlinuz-6.19.14+kali-amd64  
(belal@kali)-[/]  
$ cat ./home/belal/Defination.txt  
1- etc : System configuration  
2- Home : User home directore  
3- var/log : system and applincatooin log  
4- bin : Essential command binares  
5- tmp : Temporary fils  
(belal@kali)-[/]  
$ cat ./etc/hostname ./etc/os-release  
Completing file  
adduser.conf          firefox-esr/  
adjtime               fonts/  
alternatives/         freedts/  
apache2/              fstab  
apparmor/             fuse.conf  
apparmor.d/           gai.conf  
apt/                  geoclue/  
arp-scan/             ghostscript/  
avahi/               glvnd/  
bash.bashrc           gnome-system-tools/  
bash_completion       gophish/  
bash_completion.d/    gprofng.rc  
bindresvport.blacklist groff/  
binfmt.d/             group  
bluetooth/           group-  
ca-certificates/      grub.d/  
ca-certificates.conf  gshadow  
chatscripts/          gshadow-  
chromium/             gss/  
chromium.d/           gtk-2.0/  
cifs-utils/           gtk-3.0/  
cloud/                guymager/  
colord/               gvm/  
console-setup/        hdparm.conf  
libnl-3/  
libpaper.d/  
lightdm/  
lighttpd/  
locale.conf  
locale.gen  
localtime@  
logcheck/  
login.defs  
logrotate.conf  
logrotate.d/  
macchanger/  
machine-id  
magic  
magic.mime  
manpath.config  
matplotlib  
mime.types  
minicom/  
miredo/  
miredo.conf  
mke2fs.conf  
ModemManager/  
modprobe.d/  
perl/  
php/  
plymouth/  
polkit-1/  
postgresql/  
postgresql-common/  
powershell-empire/  
ppp/  
profile  
profile.d/  
protocols  
proxchains4.conf  
pulse/  
python2.7/  
python3/  
python3.13/  
radcli/  
rc0.d/  
rc1.d/  
rc2.d/  
rc3.d/  
rc4.d/  
rc5.d/  
rc6.d/  
ssl/  
sslsplit/  
strongswan.conf  
strongswan.d/  
stunnel/  
subgid  
subuid  
subuid-  
subversion/  
sudo.conf  
sudoers  
sudoers.d/  
sudo_logsrvd.conf  
supercat/  
sv/  
sysctl.d/  
sysstat/  
systemd/  
terminfo/  
texmf/  
theHarvester/  
tightvncserver.conf  
timidity/
```

```
Session Actions Edit View Help
(belal@kali)-[/]
$ cat ./etc/hostname ./etc/os-release
kali
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.2"
VERSION="2026.2"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"

(belal@kali)-[/]
$ cd ~
Kali Linux v...
(belal@kali)-[~]
$ ls -lar
total 148
-rw-r--r-- 1 belal belal 10882 Sep  3 18:33 .zshrc
-rw----- 1 belal belal   193 Sep  4 10:30 .zsh_history
-rw-r--r-- 1 belal belal   336 Sep  3 18:33 .zprofile
-rw----- 1 belal belal  4604 Sep  4 10:58 .xsession-errors.old
-rw----- 1 belal belal  4696 Sep  4 14:03 .xsession-errors
-rw----- 1 belal belal    49 Sep  4 12:52 .Xauthority
drwxr-xr-x 2 belal belal  4096 Sep  3 18:39 Videos
drwxr-xr-x 2 belal belal  4096 Sep  3 18:39 Templates
drwx----- 3 belal belal  4096 Sep  3 18:39 .ssh
drwxr-xr-x 2 belal belal  4096 Sep  3 18:39 Public
drwxr-xr-x 2 belal belal  4096 Sep  3 18:39 Projects
-rw-r--r-- 1 belal belal   807 Sep  3 18:33 .profile
drwxr-xr-x 2 belal belal  4096 Sep  3 18:39 Pictures
drwxr-xr-x 2 belal belal  4096 Sep  3 18:39 Music
```

=====Part B=====

```

(belal@kali)-[~]
$ mkdir nile-ops ./nile-ops/admin ./nile-ops/backup ./nile-ops/projects ./nile-ops/scratch ./nile-ops/admin/.secrets ./nile-ops/admin/policies ./nile-ops/projects/alpha ./nile-ops/project
s/beta ./nile-ops/projects/gamma ./nile-ops/projects/alpha/docs ./nile-ops/projects/alpha/logs ./nile-ops/projects/alpha/src ./nile-ops/projects/beta/docs ./nile-ops/projects/beta/src ./nile
e-ops/scratch/tmp1 ./nile-ops/scratch/tmp2

(belal@kali)-[~]
$ touch ./nile-ops/admin/contacts.txt ./nile-ops/admin/.secrets/vault.note ./nile-ops/admin/policies/backup-policy.txt ./nile-ops/projects/alpha/docs/readme.txt ./nile-ops/projects/alpha/
src/main.txt ./nile-ops/projects/alpha/src/utils.txt ./nile-ops/scratch/junk.txt ./nile-ops/scratch/.hidden_note

(belal@kali)-[~]
$ ls -ar nile-ops ./nile-ops/admin ./nile-ops/admin/.secrets ./nile-ops/admin/policies ./nile-ops/projects ./nile-ops/projects/alpha ./nile-ops/projects/alpha/docs ./nile-ops/projects/alp
ha/src ./nile-ops/projects/beta ./nile-ops/scratch
nile-ops/scratch:
tmp2 tmp1 junk.txt .hidden_note .. .
./nile-ops/projects/beta:
src docs .. .
./nile-ops/projects/alpha/src:
utils.txt main.txt .. .
./nile-ops/projects/alpha/docs:
readme.txt .. .
./nile-ops/projects/alpha:
src logs docs .. .
./nile-ops/projects:
gamma beta alpha .. .
./nile-ops/admin/.secrets:
vault.note .. .
./nile-ops/admin/policies:
backup-policy.txt .. .
./nile-ops/admin:
.secrets policies contacts.txt .. .
nile-ops:
scratch projects backup admin .. .

(belal@kali)-[~]
$ ls -l ./nile-ops/projects/alpha/src/main.txt

(belal@kali)-[~]
$ ls -l ./nile-ops/projects/alpha/src/main.txt
-rw-rw-r-- 1 belal belal 0 Sep  5 09:13 ./nile-ops/projects/alpha/src/main.txt

(belal@kali)-[~]
$ touch ./nile-ops/projects/alpha/src/main.txt

(belal@kali)-[~]
$ ls -l ./nile-ops/projects/alpha/src/main.txt
-rw-rw-r-- 1 belal belal 0 Sep  5 09:14 ./nile-ops/projects/alpha/src/main.txt

(belal@kali)-[~]
$

```

=====Part C=====

```

(belal@kali)-[~]
└─$ nano ./nile-ops/admin/contacts.txt

(belal@kali)-[~]
└─$ cat > ./nile-ops/projects/alpha/src/main.txt
alpha service -entry poin
version 1.0
status: draft
^C

(belal@kali)-[~]
└─$ cat >> ./nile-ops/projects/alpha/src/main.txt
owner: Belal^C

(belal@kali)-[~]
└─$ nano ./nile-ops/projects/alpha/src/utlils.txt ./nile-ops/projects/alpha/docs/readme.txt

(belal@kali)-[~]
└─$ cat ./nile-ops/projects/alpha/src/utlils.txt ./nile-ops/projects/alpha/docs/readme.txt >> ./nile-ops/projects/alpha/docs/alpha_project

(belal@kali)-[~]
└─$ cat defination2.txt
1- This is command > it deletes the text
2- This is command >> it continues the previous point
2- It would have deleted the text that was already there

(belal@kali)-[~]
└─$ cat ./nile-ops/projects/alpha/docs/alpha_project
This file contains utility functions.
Utilities help the program work correctly.
This is my Linux projects.
The project contains several files.
This file explains the project.

```

```

GNU nano 9.0                               ./nile-ops/projects/alpha/docs/alpha_project
1- Name: Belal Mohamed Hassn
2- ID: 200048003
3- Role: Junior Linux Administrator
4- Team Lead: Mona Rashid

./nile-ops/admin/.secrets:
vault.note

./nile-ops/admin/policies:
backup-policy.txt

./nile-ops/admin:
.secrets policies contacts.txt

```

=====Part D=====

```

(belal@kali)-[~]
$ cp ./nile-ops/admin/contact.txt ./nile-ops/backup/contacts_backup.txt

(belal@kali)-[~]
$ cp ./nile-ops/projects/alpha ./nile-ops/backup
cp: -r not specified; omitting directory './nile-ops/projects/alpha'

(belal@kali)-[~]
$ cp -r ./nile-ops/projects/alpha ./nile-ops/backup

(belal@kali)-[~]
$ cp ./nile-ops/projects/alpha/src/main.txt ./nile-ops/projects/alpha/src/utls.txt ./nile-ops/projects/beta/src

(belal@kali)-[~]
$ cp -r ./nile-ops/admin ./nile-ops/backup

(belal@kali)-[~]
$ ls -la ./nile-ops/backup/admin
.  ..  contact.txt  policies  .secrets

(belal@kali)-[~]
$ cp -r ./nile-ops/backup/alpha ./nile-ops/backup/alpha_snapshot

```

```

(belal@kali)-[~]
$ cat ./nile-ops/backup/contacts_backup.txt ./nile-ops/admin/contact.txt
1- Name: Belal Mohamed Hassn
2- ID: 200048003
3- Role: Junior Linux Administrator
4- Team Lead: Mona Rashid
1- Name: Belal Mohamed Hassn
2- ID: 200048003
3- Role: Junior Linux Administrator
4- Team Lead: Mona Rashid

(belal@kali)-[~]
$

```

====Part E=====

```

(belal@kali)-[~]
$ rmdir ./nile-ops/projects/gamma
rmdir: failed to remove './nile-ops/projects/gamma': Directory not empty

(belal@kali)-[~]
$ ls ./nile-ops/projects
alpha  beta

(belal@kali)-[~]
$ rmdir ./nile-ops/scratch
rmdir: failed to remove './nile-ops/scratch': Directory not empty

(belal@kali)-[~]
$ rm -r ./nile-ops/scratch/.hidden_note ./nile-ops/scratch/junk.txt ./nile-ops/scratch/tmp1 ./nile-ops/scratch/tmp2 ./nile-ops/scratch

(belal@kali)-[~]
$ rm -r -i ./nile-ops/projects/beta
rm: descend into directory './nile-ops/projects/beta'? y
rm: descend into directory './nile-ops/projects/beta/src'? yes
rm: remove regular file './nile-ops/projects/beta/src/main.txt'?
rm: remove regular file './nile-ops/projects/beta/src/utls.txt'? yes
rm: remove directory './nile-ops/projects/beta/src'? yes
rm: cannot remove './nile-ops/projects/beta/src': Directory not empty
rm: remove directory './nile-ops/projects/beta/docs'? yes

```

```

(belal@kali)-[~]
$ ls -la ./nile-ops/admin
.  ..  contact.txt  policies  .secrets

(belal@kali)-[~]
$ rm -r ./nile-ops/admin/contact.txt

(belal@kali)-[~]
$ cp ./nile-ops/backup/contacts_backup.txt ./nile-ops/admin

(belal@kali)-[~]
$ cat ./nile-ops/admin/contacts_backup.txt
1- Name: Belal Mohamed Hassn
2- ID: 200048003
3- Role: Junior Linux Administrator
4- Team Lead: Mona Rashid

(belal@kali)-[~]
$

```

=====Part F=====

```

(belal@kali)-[~]
$ touch ./nile-ops/admin/commands.log

(belal@kali)-[~]
$ nano ./nile-ops/admin/commands.log

(belal@kali)-[~]
$ cat ./nile-ops/admin/commands.log
====Part B====
mkdir
touch
ls -a -r
ls -l
====Part C====
nano
cat
cat >
cat >>

```

```

====Part C====
nano
cat
cat >
cat >>
====Part D====
cp
cp -r
ls -a
cat
====Part E====
rmdir
ls
rm -r
rm -r -i
ls -a
cp
cat

```



```

(belal@kali)-[~]
└─$ ls -aR ./nile-ops
./nile-ops:
. .. admin backup projects

./nile-ops/admin:
. .. commands.log contacts_backup.txt policies .secrets

./nile-ops/admin/policies:
. .. backup-policy.txt

./nile-ops/admin/.secrets:
. .. vault.note

./nile-ops/backup:
. .. admin alpha alpha_snapshot contacts_backup.txt

./nile-ops/backup/admin:
. .. contact.txt policies .secrets

./nile-ops/backup/admin/policies:
. .. backup-policy.txt

```

```

./nile-ops/backup/alpha:
. .. docs logs src

./nile-ops/backup/alpha/docs:
. .. alpha_project readme.txt

./nile-ops/backup/alpha/logs:
. ..

./nile-ops/backup/alpha/src:
. .. main.txt utils.txt

./nile-ops/backup/alpha_snapshot:
. .. docs logs src

./nile-ops/backup/alpha_snapshot/docs:
. .. alpha_project readme.txt

./nile-ops/backup/alpha_snapshot/logs:
. ..

./nile-ops/backup/alpha_snapshot/src:
. .. main.txt utils.txt

```

```

(belal@kali)-[~]
└─$ nano answers.txt

(belal@kali)-[~]
└─$ cat answers.txt
A1- pwd
A2- etc , bin , home , usr , var , tmp
A3- etc: Contains system configuratoin files
    home: Contains users home diractories
    var/log: Contains system log files
    bin: Contains essential command binaries
    tmp: Stroes temporary files created by teh system and users
A4- kali
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.2"
VERSION="2026.2"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"

```



```

B4- Absolute path
    Relative path
D2- cp: -r not specified; omitting directory
D5- If the target directory exists, cp -r places the source inside it. If it does not exist, it creates it with that name.
E2-Error message: Directory not empty
Reason: rmdir only removes empty directories, but scratch contains files.
E6- Why dangerous: It forcefully and recursively deletes files/directories without asking for confirmation, with no trash bin
to restore them.
Two safety habits:
1. Always double-check the path before pressing Enter.
2. Use the interactive flag (-i) or test with ls first to see what will be deleted.

```

```

(belal@kali)-[~]
$

```

```

(belal@kali)-[~]
$ nl answers.txt ./nile-ops/backup/admin/contact.txt ./nile-ops/admin/commands.log > ./nile-ops/submission.txt

```

```

(belal@kali)-[~]
$ nl ./nile-ops/submission.txt
1 1 A1- pwd
2 2 A2- etc , bin , home , usr , var , tmp
3 3 A3- etc: Contains system configuratoin files
4 4 home: Contains users home directories
5 5 var/log: Contains system log files
6 6 bin: Contains essential command binaries
7 7 tmp: Stroes temporary files created by teh system and users
8 8 A4- kali
9 9 PRETTY_NAME="Kali GNU/Linux Rolling"
10 10 NAME="Kali GNU/Linux"
11 11 VERSION_ID="2026.2"
12 12 VERSION="2026.2"
13 13 VERSION_CODENAME=kali-rolling
14 14 ID=kali
15 15 ID_LIKE=debian
16 16 HOME_URL="https://www.kali.org/"
17 17 SUPPORT_URL="https://forums.kali.org/"
18 18 BUG_REPORT_URL="https://bugs.kali.org/"
19 19 ANSI_COLOR="1;31"
20 20 A5- cd ~
21 21 cd -

```

```

22 22 A6- ls -la
23
24 23 B4- Absolute path
25 24 Relative path
26 25 D2- cp: -r not specified; omitting directory
27 26 D5- If the target directory exists, cp -r places the source inside it. If it does not exist, it creates it wit
h that name.
28 27 E2-Error message: Directory not empty
29 28 Reason: rmdir only removes empty directories, but scratch contains files.
30 29 E6- Why dangerous: It forcefully and recursively deletes files/directories without asking for confirmation, wi
th no trash bin to restore them.
31 30 Two safety habits:
32 31 1. Always double-check the path before pressing Enter.
33 32 2. Use the interactive flag (-i) or test with ls first to see what will be deleted.
34 33 1- Name: Belal Mohamed Hassn
35 34 2- ID: 200048003
36 35 3- Role: Junior Linux Administrator
37 36 4- Team Lead: Mona Rashid
38 37 =====Part B=====
39 38 mkdir
40 39 touch
41 40 ls -a -r
42 41 ls -l
43 42 =====Part C=====
44 43 nano
45 44 cat

```

```

45 belal 44 cat >
46 belal 45 cat >
47 belal 46 cat >>
48 belal 47 =====Part D=====
49 belal 48 cp
50 belal 49 cp -r
51 belal 50 ls -la
52 belal 51 cat
53 belal 52 =====Part E=====
54 belal 53 rmdir
55 belal 54 ls
56 belal 55 rm -r
57 belal 56 rm -r -i
58 belal 57 ls -la
59 belal 58 cp
60 belal 59 cat

(belal@kali)-[~]
$

```

=====Bonus=====

```

(belal@kali)-[~]
$ touch ./nile-ops/"team notes.txt"

(belal@kali)-[~]
$ cat ./nile-ops/"team notes.txt"

(belal@kali)-[~]
$ cd ./nile-ops/projects/alpha/src

(belal@kali)-[~/nile-ops/projects/alpha/src]
$ cd ../../../../backup

(belal@kali)-[~/nile-ops/backup]
$ ls ~/nile-ops/projects/alpha/src
main.txt  utils.txt

(belal@kali)-[~/nile-ops/backup]
$

```